Project Report

Publish–Subscribe Notification System using TCP & SSL

Team Members:
Aarohi Jaiswal - PES1UG24CS009
Abhiraj Dhananjay - PES1UG24CS016
Aaditya Vashisht - PES1UG24CS004

1. ABSTRACT

This project implements a secure publish–subscribe communication system using TCP socket programming in Python. The system allows multiple clients to connect to a server, subscribe to topics, and receive real-time notifications.

A custom application-layer protocol is designed to support commands like SUBSCRIBE, PUBLISH, and UNSUBSCRIBE. The system uses SSL/TLS encryption to ensure secure communication between clients and server.

The project demonstrates key Computer Networks concepts such as client-server architecture, TCP communication, concurrency, and secure data transmission.

2. OBJECTIVE

The main objectives of this project are:

- To implement a client-server communication model
- To design a publish–subscribe system
- To use TCP sockets for reliable communication
- To enable real-time message delivery
- To ensure secure communication using SSL/TLS
- To handle multiple clients concurrently

3. SYSTEM ARCHITECTURE

- Server acts as a message broker
- Clients act as publishers and subscribers
- Messages are routed based on topics

4. TECHNOLOGIES USED

- Programming Language: Python
- Protocol: TCP
- Security: SSL/TLS
- Concepts Used:
    - Socket Programming
    - Multi-threading
    - Application-layer Protocol Design

5. IMPLEMENTATION DETAILS

5.1 Server Side

The server is responsible for:

- Accepting multiple client connections
- Maintaining topic subscriptions
- Routing messages to subscribers
- Handling client disconnections

Key Components:

1. Socket Creation

- TCP socket created using IPv4
- Bound to port (5000)

2. SSL Configuration

- Server uses self-signed certificate (cert.pem, key.pem)
- Enables encrypted communication

3. Multi-threading

- Each client handled using a separate thread
- Ensures concurrent communication

4. Topic Management

```
topics = {
  "sports": [client1, client2]
}
```

- Stores mapping of topics to subscribers

5.2 Client Side

The client performs:

- Establishing secure connection with server
- Sending commands
- Receiving messages in real-time

Key Features:

- Uses SSL-wrapped TCP socket
- Runs a separate thread to receive messages
- Supports commands:
    - SUBSCRIBE
    - UNSUBSCRIBE
    - PUBLISH
    - PREMIUM_PUBLISH

5.3 Communication Protocol

A custom text-based protocol is used:

SUBSCRIBE <topic>
UNSUBSCRIBE <topic>
PUBLISH <topic> <message>
PREMIUM_PUBLISH <topic> <message>

6. SECURITY IMPLEMENTATION

- SSL/TLS is used to encrypt communication
- Self-signed certificates generated using OpenSSL
- Prevents data interception

7. WORKING OF THE SYSTEM

Step 1:

Client connects to server (TCP + SSL handshake)

Step 2:

Client subscribes to topic

SUBSCRIBE sports

Step 3:

Another client publishes message

PUBLISH sports Hello

Step 4:

Server forwards message to subscribers

Step 5:

Subscribers receive message instantly

8. OUTPUT / SCREENSHOTS

Server Running

```
seed@VM: ~/.../socket programming
seed@VM: ~/.../socket programming   seed@VM: ~/.../socket programming   seed@VM: ~/.../socket programming
[04/21/26]seed@VM:~/.../socket programming$ python3 server.py
[LISTENING] on 0.0.0.0:5000
[CONNECTED] ('172.20.10.6', 59398)
[CONNECTED] ('172.20.10.6', 59399)
```

Client 1 (non premium subscriber) Connects

```
seed@VM: ~/.../client
seed@VM: ~/.../client   seed@VM: ~/.../client
[04/21/26]seed@VM:~/.../client$ python3 client.py
Enter Username: user1
Username: Enter Password: 123
Password: Authentication Successful
```

Client 2 (premium subscriber) Connects

```
seed@VM: ~/.../client
seed@VM: ~/.../client    seed@VM: ~/.../client
[04/21/26]seed@VM:~/.../client$ python3 client.py
Enter Username: user2
Username: Enter Password: 123
Password: Authentication Successful
```

Client 1 Subscribes to sports

```
seed@VM: ~/.../client
seed@VM: ~/.../client    seed@VM: ~/.../client
[04/21/26]seed@VM:~/.../client$ python3 client.py
Enter Username: user1
Username: Enter Password: 123
Password: Authentication Successful
SUBSCRIBE sports
Subscribed to sports
```

Client 2 subscribes to sports

```
seed@VM: ~/.../client
seed@VM: ~/.../client    seed@VM: ~/.../client
[04/21/26]seed@VM:~/.../client$ python3 client.py
Enter Username: user2
Username: Enter Password: 123
Password: Authentication Successful
SUBSCRIBE sports
Subscribed to sports
```

```
seed@VM: ~/.../socket programming
seed@VM: ~/.../socket programming    seed@VM: ~/.../socket programming    seed@VM: ~/.../socket programming
[04/21/26]seed@VM:~/.../socket programming$ python3 server.py
[LISTENING] on 0.0.0.0:5000
[CONNECTED] ('172.20.10.6', 59398)
[CONNECTED] ('172.20.10.6', 59399)
[RECV] ('172.20.10.6', 59399): SUBSCRIBE sports
[RECV] ('172.20.10.6', 59398): SUBSCRIBE sports
```

Publisher Connected and Publishes Notification For Premium Members

```
seed@VM: ~/.../socket programming
seed@VM: ~/.../socket programming   seed@VM: ~/.../socket programming   seed@VM: ~/.../socket programming
[04/21/26]seed@VM:~/.../socket programming$ python3 client.py
Enter Username: admin
Username: Enter Password: admin
Password: Authentication Successful
PUBLISH_PREMIUM sports vip dinner with dhoni at 2:00 pm
```

Server

```
seed@VM: ~/.../socket programming
seed@VM: ~/.../socket programming   seed@VM: ~/.../socket programming   seed@VM: ~/.../socket programming
[04/21/26]seed@VM:~/.../socket programming$ python3 server.py
[LISTENING] on 0.0.0.0:5000
[CONNECTED] ('172.20.10.6', 59398)
[CONNECTED] ('172.20.10.6', 59399)
[RECV] ('172.20.10.6', 59399): SUBSCRIBE sports
[RECV] ('172.20.10.6', 59398): SUBSCRIBE sports
[CONNECTED] ('172.20.10.2', 40210)
[RECV] ('172.20.10.2', 40210): PUBLISH_PREMIUM sports vip dinner with dhoni at 2:00 pm
```

Publisher sends another notification to all subscriber on channel sports

```
seed@VM: ~/.../socket programming
seed@VM: ~/.../socket programming   seed@VM: ~/.../socket programming   seed@VM: ~/.../socket programming
[04/21/26]seed@VM:~/.../socket programming$ python3 client.py
Enter Username: admin
Username: Enter Password: admin
Password: Authentication Successful
PUBLISH_PREMIUM sports vip dinner with dhoni at 2:00 pm
PUBLISH sports match delayed for 30 min due to rain
```

Client 1 (non premium subscriber)

```
seed@VM: ~/.../client
seed@VM: ~/.../client   seed@VM: ~/.../client
[04/21/26]seed@VM:~/.../client$ python3 client.py
Enter Username: user1
Username: Enter Password: 123
Password: Authentication Successful
SUBSCRIBE sports
Subscribed to sports
[sports] match delayed for 30 min due to rain
UNSUBSCRIBE sports
Unsubscribed from sports
```

Client 2(premium subscriber)

```
seed@VM: ~/.../client
seed@VM: ~/.../client    seed@VM: ~/.../client
[04/21/26]seed@VM:~/.../client$ python3 client.py
Enter Username: user2
Username: Enter Password: 123
Password: Authentication Successful
SUBSCRIBE sports
Subscribed to sports
[PREMIUM sports] vip dinner with dhoni at 2:00 pm
[sports] match delayed for 30 min due to rain
```

Client 1 Unsubscribes

```
seed@VM: ~/.../client
seed@VM: ~/.../client    seed@VM: ~/.../client
[04/21/26]seed@VM:~/.../client$ python3 client.py
Enter Username: user1
Username: Enter Password: 123
Password: Authentication Successful
SUBSCRIBE sports
Subscribed to sports
[sports] match delayed for 30 min due to rain
UNSUBSCRIBE sports
Unsubscribed from sports
```

Server final output after both the clients disconnect

```
[04/21/26]seed@VM:~/.../socket programming$ python3 server.py
[LISTENING] on 0.0.0.0:5000
[CONNECTED] ('172.20.10.6', 59398)
[CONNECTED] ('172.20.10.6', 59399)
[RECV] ('172.20.10.6', 59399): SUBSCRIBE sports
[RECV] ('172.20.10.6', 59398): SUBSCRIBE sports
[CONNECTED] ('172.20.10.2', 40210)
[RECV] ('172.20.10.2', 40210): PUBLISH_PREMIUM sports vip dinner with dhoni at 2:00 pm
[RECV] ('172.20.10.2', 40210): PUBLISH sports match delayed for 30 min due to rain
[RECV] ('172.20.10.6', 59398): UNSUBSCRIBE sports
[DISCONNECTED] ('172.20.10.6', 59399)
[DISCONNECTED] ('172.20.10.6', 59398)
```